53.1: Biotic and abiotic factors affect population density, dispersion, and demographics

A ****__population__ is a group of individuals of a single species living in the same general area. Members of a population rely on the same resources, are influenced by similar environmental factors, and are likely to interact and breed with one another.

Populations are often described by their boundaries and size (the number of individuals living within those boundaries). Ecologists usually begin investigating a population by defining boundaries appropriate to the organism under study and to the questions being asked. A population's boundaries may be natural ones, as in the case of an island or a lake, or they may be arbitrarily defined by an investigator—for example, a specific county in Minnesota for a study of oak trees.

In this chapter, we will explore how ecologists describe and analyze populations and the factors that can determine the abundance of organisms. We will then examine recent trends in the size and makeup of the human population.

Density and Dispersion

The **density** of a population is the number of individuals per unit area or volume: the number of oak trees per square kilometer in the Minnesota county or the number of *Escherichia coli* bacteria per milliliter in a test tube. **Dispersion** is the pattern of spacing among individuals within the boundaries of the population.

Density: A Dynamic Perspective

In some cases, population size and density can be determined by counting all individuals within the boundaries of the population. We could count all the sea stars in a tide pool, for instance. Large mammals that live in herds, such as elephants, can sometimes be counted accurately from airplanes.

In most cases, however, it is impractical or impossible to count all individuals in a population. Instead, ecologists use various sampling techniques to estimate densities and total population sizes. They might count the number of oak trees in several randomly located 100×100 m plots, calculate the average density in the plots, and then extend the estimate to the population size in the entire area. Such estimates are most accurate when there are many sample plots and when the habitat is fairly homogeneous. In other cases, instead of counting single organisms, ecologists estimate density from an indicator of population size, such

as the number of nests, burrows, tracks, or fecal droppings. Ecologists also use the **mark-recapture method** to estimate the size of wildlife populations **(Figure 53.2)**.

Figure 53.2

Research Method: Determining Population Size Using the Mark-Recapture Method

Application

Ecologists cannot count all the individuals in a population if the organisms move too quickly or are hidden from view. In such cases, researchers often use the mark-recapture method to estimate population size. Andrew Gormley and his colleagues at the University of Otago applied this method to a population of endangered Hector's dolphins (*Cephalorhynchus hectori*) near Banks Peninsula, in New Zealand.

Two Hector's dolphins.

Technique

Scientists typically begin by capturing a random sample of individuals in a population. They tag, or "mark," each individual and then release it. With some species, researchers can identify individuals without physically capturing them. For example, Gormley and colleagues identified 180 Hector's dolphins by photographing their distinctive dorsal fins from boats.

After waiting for the marked or otherwise identified individuals to mix back into the population, usually a few days or weeks, scientists capture or sample a second set of individuals. At Banks Peninsula, Gormley's team encountered 44 dolphins in their second sampling, 7 of which they had photographed before. The number of marked animals captured in the second sampling (x) divided by the total number of animals captured in the second sampling (n) should equal the number of individuals marked and released in the first sampling (s) divided by the estimated population size (N):

xn=sNor, solving for population size,N=snx

The method assumes that marked and unmarked individuals have the same probability of being captured or sampled, that the marked organisms have mixed completely back into the population, and that no individuals are born, die, immigrate, or emigrate during the resampling interval.

Results

Based on these initial data, the estimated population size of Hector's dolphins at Banks Peninsula would be 180×44/7=1,131 individuals. Repeated sampling by Gormley and colleagues suggested a true population size closer to 1,100.

Data from A. M. Gormley et al., Capture-recapture estimates of Hector's dolphin abundance at Banks Peninsula, New Zealand, *Marine Mammal Science* 21:204–216 (2005).

INTERPRET THE DATA

Suppose that none of the 44 *dolphins encountered in the second sampling had been photographed before. Would you be able to solve the equation for* N*? What might you conclude about population size in this case?*

Density is not a static property but can increase or decrease over time as individuals are added to or removed from a population. As seen on the opening page of this chapter, additions occur through birth (which we define here to include all forms of reproduction) and **immigration**, the influx of new individuals from other areas. The factors that remove individuals from a population are death (mortality) and **emigration**, the movement of individuals out of a population and into other locations.

While birth and death rates influence the density of all populations, immigration and emigration can also have substantial effects. For example, studies of a population of Hector's dolphins (see Figure 53.2) in New Zealand showed that immigration accounted for about 15% of the total population size each year. Emigration was also important, typically occurring during the winter season when the dolphins move farther from shore. In general, both immigration and emigration represent key factors affecting the populations of many species. Emigration, for example, not only reduces the density of the original population, it also can result in the establishment of new populations in regions of favorable habitat, as we'll discuss later in this chapter.

Animation

Techniques for Estimating Population Density and Size

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Patterns of Dispersion

Within the boundaries of a population, the spacing among individuals may differ substantially, creating contrasting patterns of dispersion. Such differences in spacing can provide insight into the biotic and abiotic factors that affect individuals in the population.

The most common pattern of dispersion is *clumped*, in which individuals are aggregated in patches. Plants and fungi are often clumped where soil conditions and other environmental factors favor germination and growth. Mushrooms, for instance, may be clumped within and on top of a rotting log. Insects and salamanders may be clumped under the same log because of the higher humidity there. Clumping of animals may also be associated with mating behavior. Sea stars group together in tide pools, where food is readily available and where they can breed successfully **(Figure 53.3a)**. The aggregation of individuals into groups may also increase the effectiveness of predation or defense; for example, a wolf pack is more likely than a single wolf to subdue a moose, and a flock of birds is more likely than a single bird to warn of a potential attack.

Figure 53.3

Patterns of dispersion within a population's geographic range.

Patterns of dispersion within a population's geographic range.

Parts are as follows.

Part a. Clumped. Sea stars group together where food is abundant. Dot diagram: clumps of dots separated by space.

Part b. Uniform. Nesting albatrosses exhibit uniform (even) spacing, maintained by aggressive interactions between neighbors. Dot diagram: dots spaced evenly.

Part c. Random. Dandelions grow from windblown seeds that land at random and later germinate in a field scattered with dandelion flowers. Dot diagram: dots spaced with no pattern.

Patterns of dispersion can depend on scale. How might the penguin dispersion look from an airplane flying high above the ocean?

*For suggested answer, see**Appendix A*.

A *uniform*, or evenly spaced, pattern of dispersion may result from direct interactions between individuals in the population. Some plants secrete chemicals that inhibit the germination and growth of nearby individuals that could compete for resources. Animals often exhibit uniform dispersion as a result of antagonistic social interactions, such as **territoriality**—the defense of a bounded physical space against encroachment by other individuals **(Figure 53.3b)**.

In *random* dispersion (unpredictable spacing), the position of each individual in a population is independent of other individuals. This pattern occurs in the absence of strong attractions or repulsions among individuals or where key physical or chemical factors are relatively constant across the study area. Plants established by windblown seeds, such as dandelions, may be randomly distributed in a fairly uniform habitat **(Figure 53.3c)**.

Demographics

The biotic and abiotic factors that influence population density and dispersion patterns also influence other characteristics of populations, including birth, death, and migration rates. **Demography** is the study of these key characteristics of populations and how they change over time. A useful way to summarize demographic information for a population is to make a life table.

Life Tables

A **life table** summarizes the survival and reproductive rates of individuals in specific age-groups within a population. To construct a life table, researchers often follow the fate of a **cohort**, a group of individuals of the same age, from birth until all of the individuals are dead. Building the life table requires determining the proportion of the cohort that survives from one age-group to the next. It is also necessary to keep track of the number of offspring produced by females in each age-group.

Demographers who study sexually reproducing species often ignore the males and concentrate on the females in a population because only females produce offspring. Using this approach, a population is viewed in terms of females giving rise to new females. **Table 53.1** is a life table built in this way for female Belding's ground squirrels (*Urocitellus beldingi*) from a population located in the Sierra Nevada mountains of California. Next, we'll take a closer look at some of the data presented in a life table.

A squirrel eating a blade of grass.

Table 53.1

Life Table for Female Belding's Ground Squirrels (Tioga Pass, in the Sierra Nevada Mountains of California)

A table titled Table 53.1 Life table for Female Belding's Ground Squirrels (Tioga Pass, in the Sierra Nevada Mountains of California.

Table 53.1 Life table for Female Belding's Ground Squirrels (Tioga Pass, in the Sierra Nevada Mountains of California
Age (years)

0 to 1
1 to 2
2 to 3
3 to 4
4 to 5
5 to 6
6 to 7
7 to 8
8 to 9
9 to 10
Data from P.W. Sherman and M.L.Morton, Demography of Belding's ground squirrel, Ecology 65: 1617-1628(1984).
superscript asterisk: Indicates the proportion of the original cohort of 653 individuals that are still alive at the start of a time interval.
superscript dagger sign: The death rate is the proportion of individuals alive at the start of a time interval that die during that time interval.

Table 53.1 Walkthrough

Life Table for Female Belding’s Ground Squirrels

Watch an explanation of this table in the Table 53.1 Walkthrough: Life Table for Female Belding’s Ground Squirrels (Tioga Pass, in the Sierra Nevada Mountains of California)

image

Play

image

Play

Survivorship Curves

The survival rate data in a life table can be represented graphically as a **survivorship curve**, a plot of the proportion or numbers in a cohort still alive at each age. As an example, let's use the data for female Belding's ground squirrels in Table 53.1 to draw a survivorship curve. Often, a survivorship curve begins with a cohort of a convenient size—say, 1,000 individuals. To obtain the other points in the curve for the ground squirrel population, we multiply the proportion alive at the start of each year (the third column of Table 53.1) by 1,000 (the hypothetical beginning cohort). The result is the number alive at the start of each year. Plotting these numbers versus age for female Belding's ground squirrels yields **Figure 53.4**. The approximately straight line of the plot indicates a relatively constant rate of death.

Figure 53.4

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Survivorship curve for female Belding's ground squirrels.
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A graph plots number of survivors versus age for female Belding's ground squirrels.

See table next tab

The logarithmic scale on the y-axis allows the number of survivors to be visible across the entire range (2–1,000 individuals) on the graph.

*Based on this graph, what percentage of the female ground squirrels survive to be*3 years*old?*

*For suggested answer, see*Appendix A*.*

Figure 53.4 represents just one of many patterns of survivorship exhibited by natural populations. Though diverse, survivorship curves can be classified into three general types **(Figure 53.5)**. A Type I curve is flat at the start, reflecting low death rates during early and middle life, and then drops steeply as death rates increase among older age-groups. Many large mammals, including humans and elephants, that produce few offspring but provide them with good care exhibit this kind of curve.

Figure 53.5

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Idealized survivorship curves: Types I, II, and III.
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A graph plots number of survivors on a log scale from 0 to 1000 versus Percentage of maximum life span from 0 to 100.

The graph for type 1, a human, falls with increasing steepness from (0, 1000) to (100, 0). The graph for type 2, a squirrel, is a line that falls from (0, 1000) to (100, 0). The graph for type 3, an oyster, falls with decreasing steepness from (0, 1000) to (100, 0). It is near zero where x = 50.

The y-axis is logarithmic and the x-axis is on a relative scale so that species with widely varying life spans can be presented together on the same graph.

Animation

Investigating the Survivorship Curve of Oysters

image

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In contrast, a Type III curve drops sharply at the start, reflecting very high death rates for the young, but flattens out as death rates decline for those few individuals that survive the early period of die-off. This type of curve is usually associated with organisms that produce very large numbers of offspring but provide little or no care, such as long-lived plants, many fishes, and most marine invertebrates. An oyster, for example, may release millions of eggs, but most larvae hatched from fertilized eggs die from predation or other causes. Those few offspring that survive long enough to attach to a suitable substrate and begin growing a hard shell tend to survive for a relatively long time. Type II curves are intermediate, with a constant death rate over the organism's life span. This kind of survivorship occurs in Belding's ground squirrels and some other rodents, many invertebrates, lizards, and annual plants.

Many species fall somewhere between these basic types of survivorship or show more complex patterns. In birds, mortality is often high among the youngest individuals (as in a Type III curve) but fairly constant among adults (as in a Type II curve). Some invertebrates, such as crabs, may show a "stair-stepped" curve, with brief periods of increased mortality during molts, followed by periods of lower mortality when their protective exoskeleton is hard. In addition to such variation among species, survivorship curves also can differ among the populations of a single species.

In populations experiencing low levels of immigration and emigration, survivorship is one of the two key factors determining changes in population size. The other key factor that affects how population size changes over time is reproductive rate.

Reproductive Rates

As mentioned above, demographers often ignore the males and concentrate on the females in a population because only females produce offspring. Therefore, demographers view populations in terms of females giving rise to new females. The simplest way to describe the reproductive pattern of a population is to identify how reproductive output varies with the number of breeding females and their ages.

How do ecologists estimate the number of breeding females in a population? Possible approaches include direct counts and the mark-recapture method (see Figure 53.2). Increasingly, ecologists also use molecular tools. For example, scientists working in the state of Georgia collected skin samples from 198 female loggerhead turtles between 2005 and 2009. From these samples, they amplified nuclear short tandem repeats at 14 loci using the polymerase chain reaction (PCR) and produced a genetic profile for each female **(Figure 53.6)**. They then extracted DNA from an eggshell from each turtle nest on the beaches they studied and, using their database of genetic profiles, matched the nest to a specific female. This approach allowed them to determine how many of the 198 females were breeding—and how many offspring each female produced—without having to disturb the females during egg laying.

Figure 53.6

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    Using genetic profiles from loggerhead turtle eggshells to
identify which female laid the eggs.
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Genetic profiles are collected and stored in a database for females. DNA is then extracted from eggshells and compared to profiles in database to determine which females laid the eggs.

Part 1: Developing the database: Skin samples are collected from female loggerhead turtles. In a lab, DNA is extracted from each skin sample, and short tandem repeats at 14 DNA loci are amplified by PCR. For example: T A C (repeating), G T G (repeating) or T T G (repeating) all illuminate differently on PCR. A genetic profile is determined for each turtle and stored in a database.

Part 2: Comparing samples to the database: An eggshell is collected from a loggerhead turtle nest. In a lab, DNA is extracted from the eggshell, and short tandem repeats at 14 loci are amplified by PCR (as in Part I). A genetic profile is determined for each eggshell sample. The genetic profile of the eggshell is compared with an established database containing genetic profiles of adult female loggerhead turtles. A match identifies the female that laid the eggs in the nest.

VISUAL SKILLS *Use the profiles displayed in the figure to determine which breeding female laid the eggs in the nest from which eggshell sample*#74*was taken.*

*For suggested answer, see*Appendix A.*

Reproductive output for sexual organisms such as birds and mammals is typically measured as the average number of female offspring produced by the females in a given age-group. For some organisms, the number of offspring for each female can be counted directly; alternatively, molecular methods can be used (see Figure 53.6). Researchers directly counted the offspring of the Belding's ground squirrels, which begin to

reproduce at age 1 year. The squirrels' reproductive output rises to a peak at 4–5 years of age and then gradually falls off in older females (see Table 53.1).

Age-specific reproductive rates vary considerably by species. Squirrels, for example, have a litter of two to six young once a year for less than a decade, whereas oak trees drop thousands of acorns each year for tens or hundreds of years. Mussels and other invertebrates may release millions of eggs and sperm in a spawning cycle. However, a high reproductive rate will not lead to rapid population growth unless conditions are near ideal for the growth and survival of offspring, as you'll learn in the next section.